

Sekar M

Ph.D. Research Scholar

Department of Computer Science and Engineering

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 Sekar M |  Sekar M |  SK_Hari001

RESEARCH INTERESTS

Natural Language Processing, Large Language Models, Reasoning for Language Models, Multi-Hop Question Answering, Retrieval-Augmented Generation, Question Answering, Mixture-of-Experts (MoE), Multilingual and Multi-modal AI

TECHNICAL SKILLS

- **Languages & Databases:** Python, C, C++, Java, Neo4j
- **Frameworks & Libraries:** PyTorch, HuggingFace (Transformers/PEFT), LangGraph, scikit-learn, NumPy, NLTK
- **Deep Learning & LLMs:** Fine-tuning (LoRA/PEFT), Mixture-of-Experts (MoE), Graph Neural Networks (GNNs)
- **Tools & Systems:** Linux, Git, LaTeX, Docker

EDUCATION

National Institute of Technology Tiruchirappalli, Tamil Nadu, India	July 2024 – Present
Ph.D. in Computer Science and Engineering	CGPA: 8.5/10
Government College of Engineering, Bargur (Anna University), Tamil Nadu, India	July 2022 – May 2024
M.E. in Computer Science and Engineering	CGPA: 8.0/10
Varuvan Vadivelan Institute of Technology (Anna University), Tamil Nadu, India	August 2018 – May 2022
B.E. in Computer Science and Engineering	CGPA: 7.5/10

JOURNAL PUBLICATIONS

- Sekar M, Avula Karthik, Santhanavijayan A. *Do Reasoning Models Resist Knowledge Editing? Evidence from Edit Adherence in Multi-Hop Question Answering*. *Knowledge-Based Systems*, Elsevier (2026). Q1, SCIE indexed. (Communicated)

CONFERENCE PUBLICATIONS

- Sekar M, Sumit Pramod Kumar, Lakshya Patidar, Santhanavijayan A. *Better Call MoLE: A Parameter-Efficient Mixture of Legal Experts for Indian Legal Question Answering*. Workshop on Natural Legal Language Processing (NLLP @ EMNLP 2026), Budapest, Hungary, October 24–29, 2026. (Under Review)
- Sekar M, Pranava Mohan, Santhanavijayan A. *When Models Defer: Authority Framing Drives Social Compliance, Not Re-Evaluation, in LLM Multi-Agent Critique*. ACL Rolling Review (ARR), October 2026 cycle. (Under Review)
- Sekar M, Jatavath Sirisha, Santhanavijayan A. *TalkingToM: Dual-Process Cognitive Modeling for Narrative Theory of Mind Question Answering*. 5th International Conference on Speech and Language Technologies for Low-Resource Languages (SPELLL 2026), Amrita Vishwa Vidyapeetham, Coimbatore, December 17–19, 2026. (Under Review)
- Pranava Mohan, Sekar M, Santhanavijayan A. *tensorveeras@CC-MMD 2026: Cross-Cultural Misogynistic Meme Detection using a Multimodal Mixture of Experts*. ACM ICMI 2026 Grand Challenge: CC-MMD 2026. (Accepted)
- Sekar M, Sirisha Jatavath, Santhanavijayan A. *ViBrainNet: An Attention-Based CNN–Vision Transformer Ensemble for Brain Tumor Detection and Segmentation in MRI*. 2nd Congress on Intelligent Machines and Algorithms (CIMA 2026), NIT Puducherry, May 23–24, 2026. (Accepted and Presented)
- Sekar M, J. Nafeesa Begum, K. Kumar, S. Selvi. *RowColumnPrune: An Efficient Compression of Large Language Models through Row and Column Deletion*. 10th International Conference on Latest Trends in Science, Engineering and Technology (ICLTSET’24), April 2024. ISBN 978-93-6228-897-4.
- Sekar M, Chandrakanth K, Sivasakthi R, Vasanthakumar M. *A Machine Learning Approach to Dimensionality Reduction in Large Data sets*. National Conference on Computing & Information System (NCCIS’22), May 2022.

ACADEMIC EXPERIENCE

Department of CSE, NIT Tiruchirappalli
Ph.D. Research and Teaching Experience (HTRA)

Tiruchirappalli, Tamil Nadu, India
Jul'24 – Present

- Teaching Assistant, **CSPC31 – Principles of Programming Languages** Jul'26 – Present
- **Course Instructor, CSIR15 – Essentials of Computer Science** Jan'26 – May'26
- Laboratory Teaching Assistant, **CSLR62 – Web Technology Laboratory** Jan'26 – May'26
- **Course Instructor, CSHO12 – Distributed Systems** Jun'25 – Dec'25

Capgemini
Software Developer Intern

Bengaluru, Karnataka, India
Feb'22 – Jun'22

CONFERENCES & WORKSHOPS ATTENDED

Sarvam Epoch – Builder Edition
Sarvam AI

Bengaluru, India
29th July 2026

6th Indian Symposium on Machine Learning (IndoML 2025)
Graduate Forum – Poster Presentation: *TamilaAR: A Tamil Analogy Test Set for Multilingual Analogical Reasoning in Large Language Models*

BITS Pilani Hyderabad, India
Accepted

ACM Annual Event
PSG College of Technology

Coimbatore, India
1st March 2025

21st International Conference on Natural Language Processing (ICON 2024)
NLP Association of India (NLPAI) – AU-KBC Research Centre

Chennai, India
19th – 22nd December 2024

Lecture: How Could Machines Reach Human-Level Intelligence?
Speaker: Yann LeCun (VP & Chief AI Scientist, Meta) – Indian Institute of Technology Madras

Chennai, India
22nd October 2024

Build with AI
Google Developer Groups (GDG)

Bengaluru, India
23rd March 2024

Art of Computing Workshop
Department of CSA, Indian Institute of Science

Bengaluru, India
17th – 18th February 2024

ACHIEVEMENT

- **Overall 2nd Rank** in the **Cross-Cultural Misogynistic Meme Detection (CC-MMD 2026) Grand Challenge**, co-located with the ACM International Conference on Multimodal Interaction (ICMI 2026).